Direct geometric localization*

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1 A direct geometric bound

The moment argument can be made without a succession of feedback updates. The useful quantity is a single cost for changing the sign probability, moving the inner support endpoints, and adding conditional noise. We prove its scalar consequences directly, then evaluate it once on each of the already available probability intervals.

Let X have mean zero, variance one and compact support with an actual central gap $(-A, B)$, where $A, B > 0$ are support endpoints. Set

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_-, \quad \bar{A} = m/q, \quad \bar{B} = m/p,$$

$$\theta = 1 - m^2/v, \quad w = 1 - \theta, \quad \ell = m/v = \sqrt{w/v}, \quad d = 1 - (A + B)/\ell.$$

There is no atom at zero. Conditional means give $A \le \bar{A} = \ell p$, $B \le \bar{B} = \ell q$, so $0 \le d < 1$. Write $W = X - \mathbb{E}[X \mid \mathbf{1}_{X>0}]$ and define measures on magnitudes by

$$\int f \, d\mu_\pm = \mathbb{E}[W^2 \mathbf{1}_{\{\pm X > 0\}} f(|X|)].$$

Their total mass is θ. For $\sigma = \sqrt{10}$ put

$$Q(p) = (\sigma + 1)p^2 + (\sigma - 1)q^2 - 6pq = (6 + 2\sigma)(p - p_*)^2, \qquad p_* = (4 - \sigma)/2.$$

Let $\rho = \mathbb{E}|X|^3$, $\gamma = \mathbb{E}X^3$, $\beta = \mathbb{E}[X|X|]$, $J = \sum_{(a,b)} \tau_{a,b}(b - a)^2$, and $D = (3 + \sigma)\rho - \gamma - 3J$. The sum is over bounded support gaps and $\tau(t) = \mathbb{E}[X \mathbf{1}_{X \ge t}]$ is constant on each gap.

Proposition 1.1 (One common geometric cost)**.** *With the preceding definitions,*

$$D \ge \ell\{wQ(p) + 6wd(1 - d) + \theta K(p, d)\}, \tag{1}$$

where

$$K(p, d) = \min\{3(\sigma + 1)q - (\sigma + 5)d, 3(\sigma + 3)p - (\sigma + 7)d\}.$$

No maximizing-contact or predecessor hypothesis is needed for this geometric inequality.

Proof. For the positive outer gaps,

$$J_o^+ \le 2 \int_B^\infty (t - B)\tau(t) \, dt = \mathbb{E}[X_+(X_+ - B)^2],$$

*Research proof developed with substantial AI assistance. Its numerical statements retain the stated entry and analytic hypotheses; it is not a proof of the full sharp Berry–Esseen conjecture.

and the same inequality holds on the negative side. Endpoint atoms are included; isolated integration endpoints have zero Lebesgue measure. Add the central gap contribution $m(A+B)^2$. Substitution gives

$$D \ge \ell w Q(p) + E_0 + \int k_+ \, d\mu_+ + \int k_- \, d\mu_-,$$

$$E_0 = 6m\{(A+B)(\bar{A} - A + \bar{B} - B) + (\bar{A} - A)(\bar{B} - B)\},$$

$$k_+(x) = (\sigma - 1)(x + 2\bar{B}) + 6B, \qquad k_-(x) = (\sigma + 1)(x + 2\bar{A}) + 6A.$$

For completeness, the substitutions use $\mathbb{E}X_+^3 = m\bar{B}^2 + \int (x + 2\bar{B}) d\mu_+$ and its negative analogue, together with $\mathbb{E}X_+^2 - mB = \mu_+(\mathbb{R}) + m(\bar{B} - B)$. Put $a_0 = (\bar{A} - A)/\ell$, $a_1 = (\bar{B} - B)/\ell$. Then $a_0, a_1 \ge 0$, $a_0 + a_1 = d$, and

$$E_0/\ell = 6w\{d(1-d) + a_0 a_1\}.$$

On the positive measure $x \ge B$, so $k_+(x)/\ell \ge 3(\sigma+1)q - (\sigma+5)a_1$. On the negative measure the corresponding lower bound is $3(\sigma+3)p - (\sigma+7)a_0$. Drop $a_0 a_1$, use $a_i \le d$, and sum the two nonnegative measures. Their combined mass is θ, proving the result. □

1.1 Direct global localization

The following numerical premises are the coarse entry's global output. They are not conclusions of the older moment iteration:

$$p \in [.24, .54], \quad n \ge 42, \quad nD \le 32.20665378, \qquad |X| \le 157/50. \tag{2}$$

Only the final third-moment bound uses the support radius. Define

$$D_0 = .76682509, \quad s = 3.162277660, \quad s^+ = 3.162277661, \quad z = (s - 1 - D_0)/3,$$

$$T = 7/125 = .056, \quad W_0 = 1 - T = .944, \quad d_0 = 17/250 = .068.$$

All displayed decimals in this section denote exact rationals.

Theorem 1.2. *Under* (2),

$$p > 1/4, \qquad \theta < .056, \qquad d < .068, \qquad \lambda := n\theta/\ell^2 < .616, \qquad \rho < 1.426.$$

The numerical part reduces to four quartics, two cubics and elementary rational inequalities specified below.

Proof. The preceding outer-gap bound also gives $J \le \rho + 2mAB$. Thus

$$D \ge \sigma\rho - \gamma - 6mAB \ge \sigma - 1 - 3AB,$$

using $\rho \ge 1$, $\gamma \le \rho$, $m \le 1/2$. Consequently

$$AB \ge z > 0, \quad w \ge AB \ge z, \qquad 0 \le d \le 1 - 2\sqrt{zv/w} =: d_{\max}. \tag{3}$$

Here $AB \le \bar{A}\bar{B} = w$ and $A + B \ge 2\sqrt{AB}$.

First, $\theta < 1/5$ follows without iteration. Since $B \ge z/\bar{A}$, the positive coefficient in the common budget satisfies

$$k_+/\ell \ge q\{(\sigma+5)z/w + 2(\sigma-1)\} \ge 3.13.$$

The negative coefficient is bounded by $p\{(\sigma+7)z/w + 2(\sigma+1)\} \ge 3.13$ in the same direction. These last bounds use $w \le 1$, $q \ge .46$, $p \ge .24$. Therefore $D \ge 3.13\ell\theta$; since $\ell \ge 2\sqrt{z}$, a value $\theta \ge 1/5$ would give $D \ge (313/250)\sqrt{z} > D_0$.

Suppose next $T \le \theta < 1/5$. For fixed p, θ, each branch of (1) is concave in d. It suffices to exclude $d = 0$ and $d = d_{\max}$. At the latter endpoint put $g = 1 - d_{\max} = 2\sqrt{zv/w}$. Exact rational square comparisons give $.59 < g < .77$, $g(1-g) > .17$ and $\ell > 1.7$. Both noise coefficients are positive, since $3(\sigma+1)q, 3(\sigma+3)p > 4.4$ and $4.4 - (s^+ + 7) \cdot .41 > 0$. Hence the inward cost alone exceeds $1.7 \cdot 6 \cdot .8 \cdot .17 > D_0$. At $d = 0$ the two branches are

$$F_C(\theta) = \sqrt{(1-\theta)/v}\{(1-\theta)Q(p) + \theta C(p)\},$$

where $C(p) = 3(\sigma+1)q$ or $3(\sigma+3)p$. Their derivative has the sign of $(2-3\theta)C - 3(1-\theta)Q$, which is positive because $Q \le .4$, $C > 4.4$, $\theta \le .2$. It remains to prove $F_C(T) > D_0$.

Here and below replace σ by its rational lower bound s in Q, C, writing Q_s, C_s. The required positive-base squared inequalities are

$$P_\pm(p) = W_0\{W_0 Q_s(p) + TC_{s,\pm}(p)\}^2 - D_0^2 p(1-p) > 0.$$

They have $P_\pm'' \ge 2D_0^2$ on $[.24, .54]$. If a twice differentiable function satisfies $P'' \ge 2a > 0$, then

$$P(p) \ge P(x) - P'(x)^2/(4a). \tag{4}$$

This follows by integrating the second derivative and completing the square. The exact evaluations at $x = .45, .39$ give respectively lower bounds .0017 and .011. This excludes both endpoints and proves $\theta < T$.

From (3), $d < .42$. Each branch of the direct bound now increases in d, since $6W_0(1 - 2 \cdot .42) - T(s^+ + 7) > 0$. At $d = d_0$, it also increases in θ: the derivative numerator is

$$(2-3\theta)(C - (\sigma+5 \text{ or } \sigma+7)d_0) - 3(1-\theta)\{Q + 6d_0(1-d_0)\} > 0.$$

The least value is therefore at $\theta = 0$. The remaining inequality is

$$P_d(p) = \{Q_s(p) + 6d_0(1-d_0)\}^2 - D_0^2 p(1-p) > 0.$$

Again $P_d'' \ge 2D_0^2$; formula (4) at .424 gives a lower bound .0012. Thus $d < d_0$.

Since the direct bound increases in d, it gives $D \ge \ell\{wQ + \theta C\}$ for at least one of the two values of C. Multiplying by n and using $n\theta = w\lambda/v$ yields

$$\lambda \le \max\left\{\frac{p[17.558 - 42Q(p)]}{3(\sigma+1)}, \frac{q[17.558 - 42Q(p)]}{3(\sigma+3)}\right\}.$$

Here $\ell w \ge 2W_0^{3/2}$ and $32.20665378/(2W_0^{3/2}) < 17.558$. Replacing Q, σ by their lower enclosures gives rational cubics U_+, U_-. The positive cubic increases up to $p = 1/3$ and satisfies $U_+'' \le -13$ thereafter. The negative cubic has $U_-'' \le -12$ on the entire interval. The concave counterpart of (4), evaluated at .455 and .392, gives $U_+ < .615289$, $U_- < .565358$, proving the stated bound on λ.

For the third moment, $\ell > 1.94$, $B > 1.94(.46 - .068)$ and $\bar{B} < 1.8$. Thus, for $x \le 3.14$,

$$\frac{k_+(x)}{x + 2\bar{B}} \ge s - 1 + \frac{6 \cdot 1.94(.46 - .068)}{3.14 + 3.6} > 5/2;$$

the negative ratio is already at least $\sigma + 1 > 5/2$. The exact third-moment identity and common budget imply

$$\rho \le \frac{w^{3/2}}{\sqrt{v}}\{1 - 2v - \tfrac{2}{5}Q(p)\} + \tfrac{2}{5}D_0 \le \frac{1 - 2v - \frac{2}{5}Q_s(p)}{\sqrt{v}} + \tfrac{2}{5}D_0.$$

The bracket is positive since $Q \le .4$. For $M = 1.426 - \frac{2}{5}D_0$, the final quartic is

$$P_\rho(p) = M^2 p(1-p) - \{1 - 2p(1-p) - \tfrac{2}{5}Q_s(p)\}^2.$$

It satisfies $P_\rho'' \ge .4$ on the full interval, and (4) at .258, with $a = .2$, gives $P_\rho > .00015$. Positivity of both squared bases proves $\rho < 1.426$. Finally $D \ge Q(p)/\sqrt{pq}$ follows by the already proved monotonicities in d and θ. This ratio decreases on $[.24, .25]$, and its value at .25 exceeds D_0 by an exact squared comparison. Hence $p > 1/4$. □

Finite arithmetic. The four quartics are specified by their displayed formulas, not by fitted coefficients. Their second derivatives are quadratic, whose minima on a closed interval occur at endpoints or at the eligible vertex. Cubic second derivatives are affine. Thus the stated derivative bounds and tangent values require only rational evaluations. The accompanying pure-Python checker reconstructs the polynomials and performs these operations with `Fraction`. It also checks positivity before squaring and the rational enclosures $s^2 < 10 < (s^+)^2$. These checks prove precisely the scalar inequalities used above, conditional on the analytic derivation.

1.2 One evaluation on each probability interval

The global lemma also removes the repeated updates needed to obtain the existing support strips. Let a closed entry panel be $I = [l, u]$, with $D \le D_I$ and $nD \le 42D_I$. Put v_-, v_+ for its actual minimum and maximum of $p(1-p)$, including $1/4$ when the panel crosses $1/2$. Let $Q_I = \max(0, \min_I Q_s)$ and

$$C_I = \min\{3(s+1)(1-u), 3(s+3)l\}, \qquad e_0 = \sqrt{W_0/v_+}.$$

Since $D \ge \ell\{Q + \theta(C-Q)\}$ and $\ell \ge e_0$, a negative $D_I/e_0 - Q_I$ excludes the whole panel. Otherwise one evaluation gives

$$\Theta_I = \min\left(T, \frac{D_I/e_0 - Q_I}{C_I - Q_I}\right), \quad e_I = \sqrt{(1-\Theta_I)/v_+}, \quad J_I = \frac{D_I}{e_I(1-\Theta_I)} - Q_I, \tag{5}$$

$$d_I = \min\left(d_0, \frac{1-\sqrt{1-2J_I/3}}{2}\right), \quad A_I = e_I(l-d_I), \quad B_I = e_I(1-u-d_I).$$

All roots and divisions are checked on their stated domains. If $J_I < 0$ the panel is impossible; otherwise the smaller quadratic root is correct because the global result has already proved $d < .068 < 1/2$. These inequalities follow from $D \ge \ell w\{Q + 6d(1-d)\}$, since the noise coefficients remain positive.

For a coherent third-moment estimate define

$$\bar{A}_u = \sqrt{u/(1-u)}, \quad \bar{B}_u = \sqrt{(1-l)/l}, \quad L_I = R + 2\max(\bar{A}_u, \bar{B}_u),$$

$$k_I = \min\{s - 1 + 6B_I/(R + 2\bar{B}_u),\ s + 1 + 6A_I/(R + 2\bar{A}_u)\}, \quad r_I = D_I - e_I(1-\Theta_I)Q_I.$$

The noise term in ρ is at most $\min(L_I\theta, r_I/k_I)$. Write $r_B = (1-2v_-)/\sqrt{v_-}$ and $\alpha_I = (3/2)\sqrt{1-\Theta_I}$. Integrating the derivative of $(1-\theta)^{3/2}$ gives the affine bound $1 - \alpha_I\theta$, which stays positive since $\alpha_I\theta \le (3/2)T < 1$. Hence

$$\rho \le \min\left(1.426,\ r_B + (L_I - \alpha_I r_B)_+ \min\{\Theta_I, r_I/(k_I L_I)\}\right).$$

This is the maximum of two affine pieces meeting at their turnover; it does not combine incompatible allocations of the same noise. The other exports are

$$1 - 2u - 2(1-u)\Theta_I \le \beta \le 1 - 2l + 2l\Theta_I, \qquad e_I \le \ell \le 1/\sqrt{v_-},$$

$$\lambda \le \frac{v_+}{C_I}\left\{\frac{42D_I}{e_I(1-\Theta_I)} - 42Q_I\right\}.$$

The same calculation retains a joint displacement–noise constraint. Set

$$a_I^* = 6(1-d_I) - \frac{\Theta_I}{1-\Theta_I}(s^+ + 7) > 0.$$

In the direct budget, $6wd(1-d) - \theta(\sigma+5 \text{ or } \sigma+7)d \geq wa_I^* d$. Multiply by n, use $n\theta = w\lambda/v$, $n \geq 42$ and $\ell w \geq e_I(1-\Theta_I)$. All remaining costs are nonnegative, so

$$a_I^* d + \frac{C_I \lambda}{42 v_+} \leq J_I. \tag{6}$$

If $J_I = 0$ this forces $d = \lambda = 0$; otherwise it is a single shared budget with intercepts J_I/a_I^* and $42v_+J_I/C_I$. It does not permit spending the full reserve independently on displacement and noise. All these are direct formulas; no result is fed back into an earlier formula.

Proposition 1.3 (Connection with the current entry)**.** *For the thirty closed nonempty panels of the selected v7 entry, the preceding single evaluation excludes panels 24 and 53 and retains panels 25 through 52. Under the fixed-target contact and sharp-predecessor hypotheses of the retained heat/ray lemma, outward rational evaluation supplies all thirteen existing $|U| < .6$ strips on $p \in [.40, .53]$ through the retained all-threshold heat and ray inequalities. On $[.41, .42]$ it gives*

$$\theta \leq .049086874003, \quad d \leq .066505471827, \quad \rho \leq 1.204660335428, \quad \lambda \leq .526219192656,$$

$$1.975749037978 \leq \ell \leq 2.033209198406, \qquad .103059226156 \leq \beta \leq .220251236683.$$

These imply every outward moment, noise and scale premise used by the current central-panel confinement theorem. The raw-threshold and signed phase bounds still come from their original separate predecessors.

Proof. The direct formulas above prove the moment and scale bounds on each entire closed panel. The finite check reads only the named entry's frozen input, retains all thirty nonempty panels, and uses directed rational square roots. For the strip test put $t_- = e_I(l + .6)$, $t_+ = e_I(1 - u + .6)$, with r the proved rho cap and $b = -\beta_l$ or $b = \beta_u$, respectively. The retained all-threshold contact lemma bounds the contact expression at $t > \sqrt{3}$ by

$$\frac{2}{5}S^3\left\{\frac{t^3-3t}{6} + \frac{|t^4-6t^2+3|}{72}\right\} + S^3\left\{\frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2-2)}{78720}\right\} + c_0 Q_r(t),$$

where $S = 81/80$, $c_0 = 819/2000$, $e = 8/875$ and $Q_r(t) = -t^3 + 3rt^2/2 + 3bt + er$. At both boundaries the finite check requires this expression negative and

$$Q_r, Q_r', Q_r'' < 0, \qquad L = \tfrac{2}{5}(t^2/2 + 1/2 + e) + c_0 Q_r' < 0, \quad L', L'' < 0.$$

These are the existing whole-ray sufficient conditions, with their original index, contact and predecessor hypotheses. Both sides pass exactly on panels 40 through 52. All thirteen closed intervals and both endpoints are retained; the other fifteen panels are not claimed to satisfy these strip tests. The displayed central vector consists of outward bounds from its single rational evaluation; direct comparisons with the previous central theorem's premises give the asserted connection. □

What this replaces. The eight-round moment iteration is unnecessary for the four global bounds, the probability exclusions, the thirteen incoming strips and the central-panel input vector just listed. The actual v7 entry, its support radius and weighted nD bound, the common gap budget and the heat/ray lemma remain dependencies. This is not a proof of the remaining high-noise panels, the count comparison or the finite base. Any downstream calculation using individual old panel bounds requires a new input-by-input check; the global improvements alone do not authorize such substitutions.